



Module 6 : Generative AI

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Agenda - Part : III

- Introduction to Generative AI
- Types of Generative AI Models
 - Variational AutoEncoders,
 - Generative Adversarial Networks
- Advantages and limitations of Generative AI
- ChatGPT and Conversational AI

Introduction to Generative AI

Definition: A branch of AI that creates **new** data (text, images, code, audio) that resembles the training data, rather than just classifying or predicting existing data.

Discriminative vs. Generative:

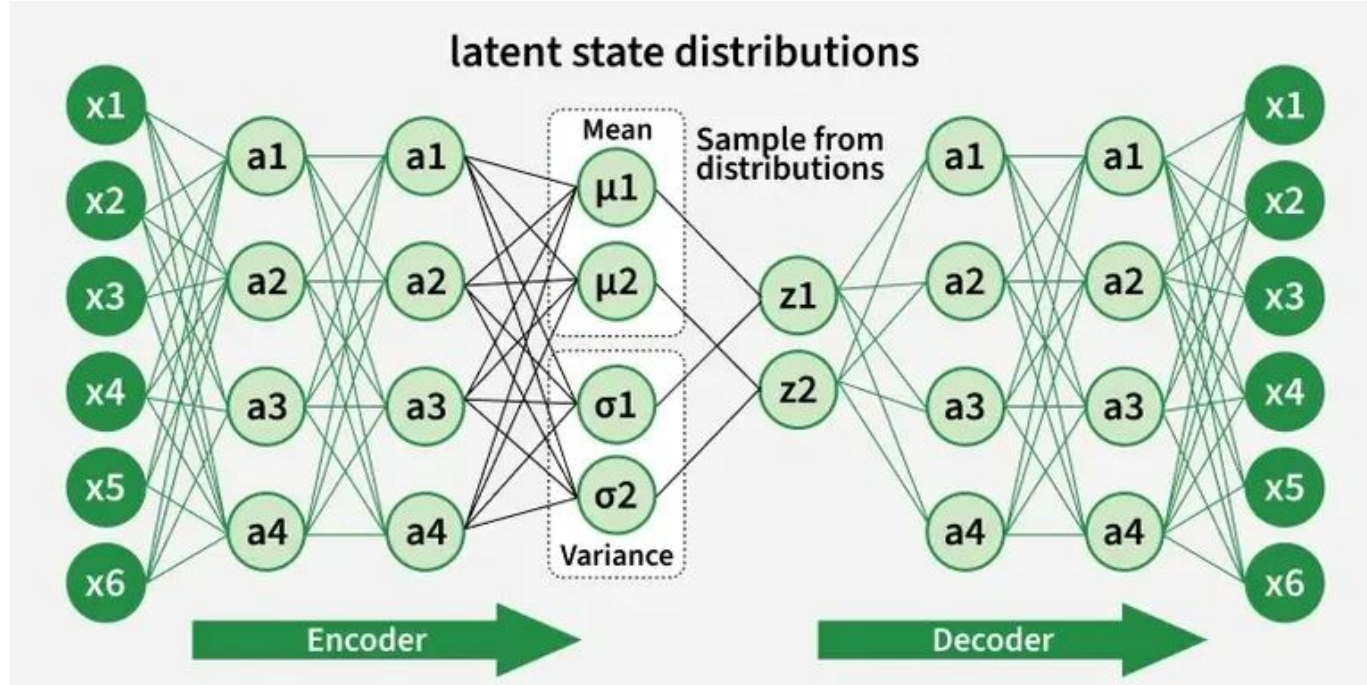
- *Discriminative:* Learns the boundary between classes (Is this a cat or a dog?).
- *Generative:* Learns the distribution of the classes (What does a cat look like?).

The Core Mechanism

- **Pattern Recognition:** GenAI identifies statistical patterns in massive datasets.
- **Probability:** It predicts the "most likely" next element (token in text, pixel in images).
- **Training:** Uses Deep Neural Networks to map high-dimensional data into a manageable mathematical space.

Types of Models – VAEs & GANs

- **Variational Autoencoders (VAEs)**





Variational Autoencoders (VAEs)

Architecture: Encoder-Decoder.

The Process:

1. **Encoder:** Compresses input data into a "Latent Space" (a lower-dimensional representation).
2. **Latent Space:** Instead of a single point, it represents data as a **probability distribution** (mean μ and variance σ).
3. **Decoder:** Samples from this distribution to reconstruct the original data.

The VAE Loss Function

To ensure the model learns well, it minimizes:

$$L = L_{reconstruction} + D_{KL}(q(z|x)||p(z))$$

Reconstruction Loss: Ensures the output looks like the input.

KL Divergence: Ensures the latent space is structured and continuous (no "gaps" in the AI's imagination).

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Generative Adversarial Networks (GANs)

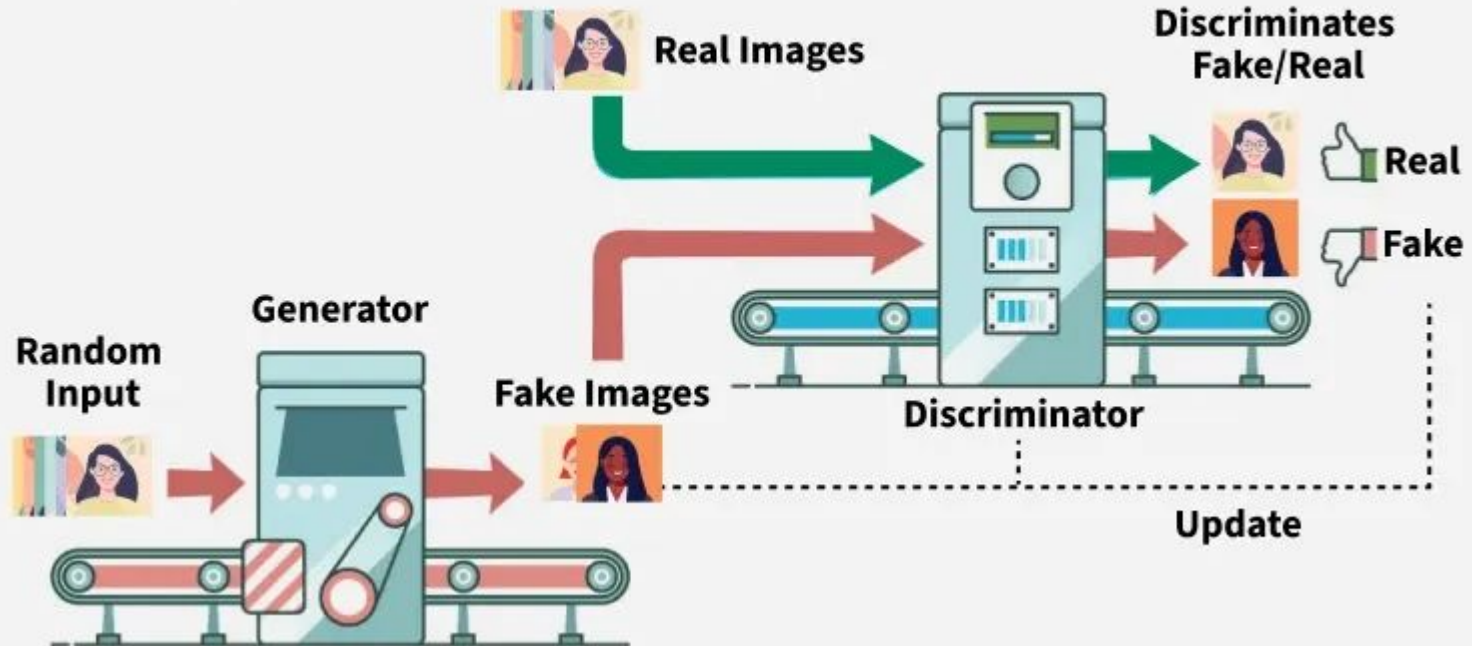
The Concept: Two neural networks competing in a "cat-and-mouse" game.

The Players:

1. **The Generator:** Creates "fake" data to try and trick the Discriminator.
2. **The Discriminator:** Tries to distinguish between "real" (training) data and "fake" data.

Outcome: Through competition, the Generator becomes so good that the Discriminator can no longer tell the difference.

Generative Adversarial Network (GANs)



GANs vs. VAEs – Comparison

Feature	VAE	GAN
Output Quality	Often blurry/smooth	Sharp and highly realistic
Training Stability	Stable and predictable	Difficult/Unstable (Mode Collapse)
Latent Space	Well-structured/Continuous	Less structured
Best Use Case	Signal analysis, Anomaly detection	Image/Video generation

Advantages & Limitations

Advantages of GenAI

- **Efficiency:** Automates tedious content creation (emails, basic code, summaries).
- **Personalization:** Can tailor content to specific user needs in real-time.
- **Creative Augmentation:** Acts as a "co-pilot" for artists and engineers, providing drafts and inspiration.

Advantages & Limitations

Critical Limitations

- **Hallucinations:** The model confidently provides incorrect information because it prioritizes "probability" over "truth."
- **Bias:** If the training data contains prejudices (racism, sexism), the AI will replicate and amplify them.
- **Data Privacy:** Entering sensitive company data into public models can lead to leaks.
- **Environmental Cost:** Training large models requires massive electrical power and water for cooling data centers.



ChatGPT & Conversational AI

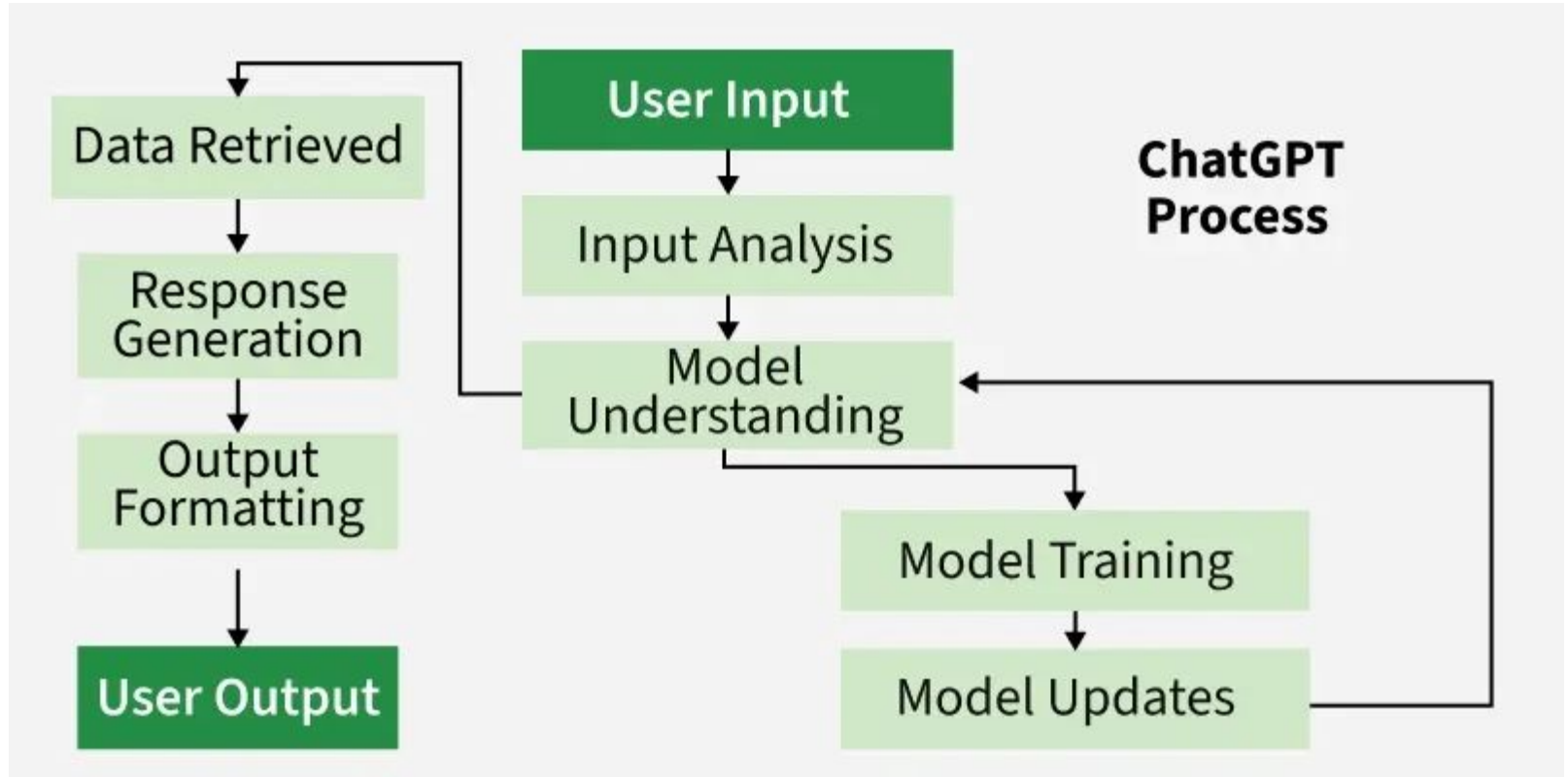
- **Architecture:** ChatGPT is based on the **Transformer** model.
- **Self-Attention:** The model can focus on different parts of a sentence to understand context (e.g., in "The bank of the river," it knows "bank" isn't a financial institution).
- **Parallelization:** Unlike older models, Transformers can process entire sentences at once, making them much faster to train.



How ChatGPT is Trained

1. **Pre-training:** Reading the entire internet to learn grammar, facts, and reasoning.
2. **Fine-tuning:** Specialized training on high-quality datasets.
3. **RLHF (Reinforcement Learning from Human Feedback):**
Humans rank the model's answers, teaching it to be helpful, honest, and harmless.

ChatGPT Process



Limitations of ChatGPT

- **Accuracy:** It can sometimes provide inaccurate or misleading information. It's important to double-check critical information.
- **Real-Time Data:** It doesn't have access to current events or real-time data as it's trained on historical data.
- **Occasional repetition:** Sometimes ChatGPT might generate repetitive responses or miss nuanced details.
- **Contextual Understanding:** The model can sometimes struggle with maintaining context during long conversations or with ambiguous inputs.
- **Biases:** Like any AI model, it may inadvertently generate biased content based on its training data.



Conversational AI

- **Multimodal:** Modern agents can see, hear, and speak (GPT-5 series integration).
- **Reasoning Tiers:** Distinction between "Instant" models (fast chat) and "Thinking" models (deep problem-solving).
- **Agentic Shift:** ChatGPT is moving from a "Chatbot" to an "Agent" that can click buttons and use tools on your behalf.